Programs on Problem Solving

a: Solve the 8 Queens Problem

Aim

To place 8 queens on an 8x8 chessboard so that no two queens attack each other, i.e., no two
queens share the same row, column, or diagonal.

Procedure:

1. Create an 8x8 board initialized with 0s.
2. Place a queen in each row by checking if it's safe.
3. If no column is safe in a row, backtrack.
4. Repeat until all 8 queens are placed.



```python
N = 8
board = [[0 for _ in range(N)] for _ in range(N)]

def is_safe(row, col):

    for i in range(row):
        if board[i][col] == 1:
            return False

    i, j = row - 1, col - 1
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1
    i, j = row - 1, col + 1
    while i >= 0 and j < N:
        if board[i][j] == 1:
            return False
        i -= 1
        j += 1

    return True

def solve(row):

    if row == N:
        return True

    for col in range(N):
        if is_safe(row, col):
            board[row][col] = 1

            if solve(row + 1):
                return True

            board[row][col] = 0

    return False

if solve(0):
    print("Solution for 8 Queens Problem:\n")

    for row in board:
        print(" ".join("Q" if cell == 1 else "." for cell in row))
else:
    print("No solution exists.")
```